



University of Pisa

Master's Degree in Computer Science

Curriculum: Artificial Intelligence



Machine Learning for Satellite Link Fade Nowcasting

Giuseppe Gabriele Russo

Experimental thesis carried out at M.B.I. S.r.l.

Supervisor
Davide Bacciu

Co-supervisor
Giovanni Scognamiglio

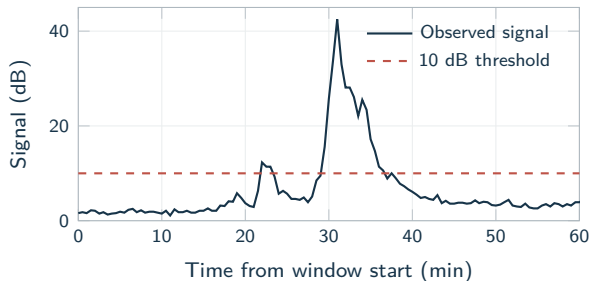
Industrial Context and Operational Problem

Experimental work at M.B.I.

Real satellite fade and beacon recordings from the Fucino ground station.

The operational question

Will the fade persist long enough to justify a backup-link switch?

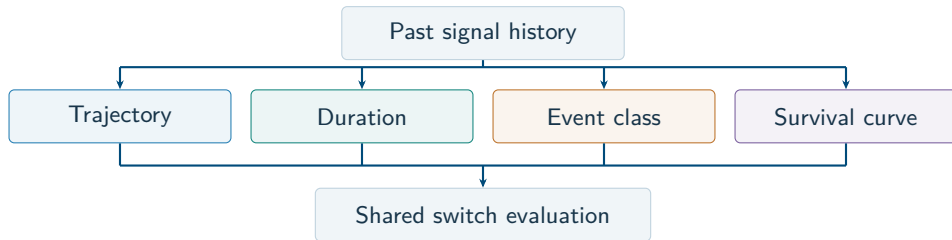


Except from the held-out source; future samples are unavailable to the predictor at decision time.

Anticipate persistence, rather than only detect a threshold crossing.

Research Objective and Contribution

Which predictive formulation supports useful switch decisions from signal history alone?



- Compare four task formulations using established model families.
- Separate predictive accuracy from operational switch behavior.
- Make decision coverage explicit when comparing different tasks.

Data and Experimental Setting

Signal-only inputs 30-second sampling Main context: 30 observations

Why sample around fades?

Fade episodes are sparse; most samples lie near the non-fade baseline.

Uniform sampling risks letting long quiet plateaus dominate the training loss.

Event-focused sample selection

Forecasting / current-level: windows extending **90 min before and after** candidate onset.

Long-fade / survival: **within-event samples**.

Development sources

Chronological training / validation splits.

Validation-based model selection.

External test source

`fc-uplink-fade.csv`

Excluded from fitting and hyperparameter search.

Event-focused evaluation is not continuous monitoring of the complete time series.

Four Predictive Formulations

Formulation	Question at time t	Output
Autoregressive	What are the next ten signal values?	Future trajectory
Current-level	How long until the signal falls below the threshold $S_t - 0.5$ dB?	Duration estimate
Long-fade	Does the entire grouped event last at least 300 seconds?	Class probability
Survival	Will the current fade remain active for more than u seconds?	Survival curve $S(u X_t)$

All use past signal information; their targets are not interchangeable.

Perfect Switch and Evaluation

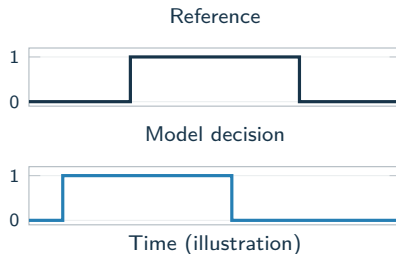
Perfect Switch

An offline reference using the true signal and future persistence.

An evaluation oracle, not a deployable predictor.

Complementary measurements

- Precision, recall, F1, balanced accuracy.
- Active duration and switch intervals.
- Timing and missed activity.



Measure what the decision does, not only how accurate the prediction is.

Shared Switch Post-Processing

1. Raw decision

Apply the task-specific conversion rule.

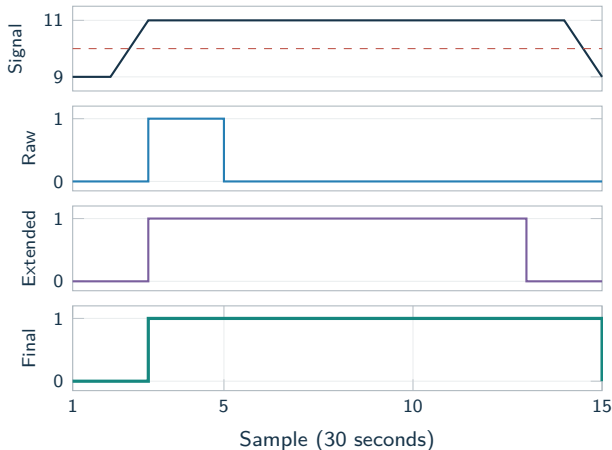
2. Minimum active interval

Extend short positive islands to the configured ten-sample length.

3. Hold while above threshold

Keep an already active switch on while the current signal remains ≥ 10 dB.

Modifies the binary decision, not the input signal.



Autoregressive Forecasting

GRU S2V / encoder-decoder PatchTST XGBoost Chronos

Predict

30 past values → **10 future values.**

Trainable models: raw and context-standardized inputs. Chronos: raw, zero-shot.

Initial switch

All ten predictions are ≥ 10 dB.

Forecast error vs. switch quality

Model	RMSE (dB) ↓	Switch F1 ↑
GRU S2V raw	2.852	0.674
PatchTST raw	3.088	0.808

External test, native L30 grid.

Lower forecast error does not guarantee better switches.

Current-Level Persistence

XGBoost Shapelets with MLP, convolutional or Transformer heads

Predict

Time until the signal falls below its
current level minus 0.5 dB.

Input: relative-to-current history plus scalar
signal features.

Initial switch

Predicted duration ≥ 5 min
and current signal ≥ 10 dB.

XGBoost: duration vs. decision

Duration MAE ↓	586.9 s
----------------	----------------

Duration median AE ↓	27.2 s
----------------------	--------

Switch F1 ↑	0.902
-------------	--------------

External test, native grid.

Shapelet variants: no final switch activations on
this test set.

Useful switches despite large errors; level recovery is not fade recovery.

Long-Fade Detection

XGBoost TCN Shapelet convolution classifier

Predict

Probability that the **whole grouped event** lasts at least 5 min.

The label stays the same throughout the event, even near its end.

Initial switch

Predicted probability ≥ 0.5 .

TCN: classification vs. decision

Classification AUPRC $\uparrow \approx 1.000$

Switch precision $\uparrow$ **0.540**

Switch recall $\uparrow$ **1.000**

External test, native grid. Event-conditioned, highly imbalanced classification.

Total event duration is not remaining duration.

Survival Persistence

XGBoost-AFT: time distribution

TCN: discrete hazards

Remaining fade duration

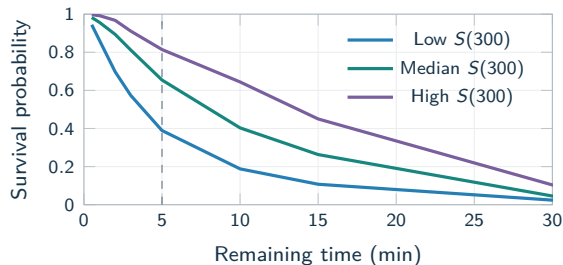
$$\mathcal{S}(u \mid X_t) = P(R_t > u \mid X_t)$$

Right-censored samples are retained.

Initial switch

Five-minute probability ≥ 0.65
and current signal ≥ 10 dB.

0.65 is exploratory/test-selected, not
validation-calibrated.



Three test TCN predictions; dashed line: 5 min.

Direct residual-persistence estimates, with calibration still required.

Cross-Task Switch Performance

Selected representatives on the common reference grid

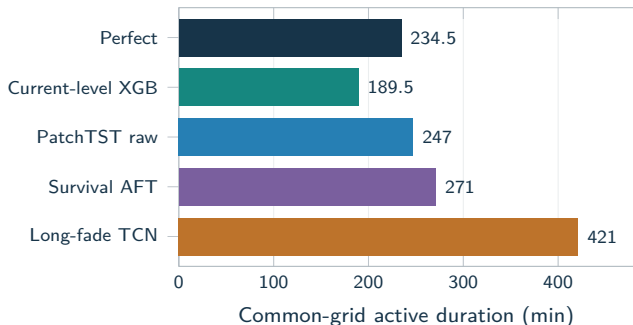
Representative	Coverage	Precision ↑	Recall ↑	F1 ↑
Current-level XGBoost	89.0%	1.000	0.808	0.894
Survival XGBoost-AFT	9.2%	0.812	0.938	0.870
PatchTST raw	89.2%	0.779	0.821	0.800
Long-fade TCN	28.7%	0.557	1.000	0.715

Missing native decisions map to zero. Coverage measures the fraction of reference timestamps with a native decision.

Survival threshold: exploratory 0.65. Descriptive test comparison, not a validation-selected cross-task winner.

Conservative activation, greater recovery and broad recall are different trade-offs.

Active Duration and Switching Events



Duration is not fragmentation

The same active time can be split into very different numbers of switch intervals.

Within-task example

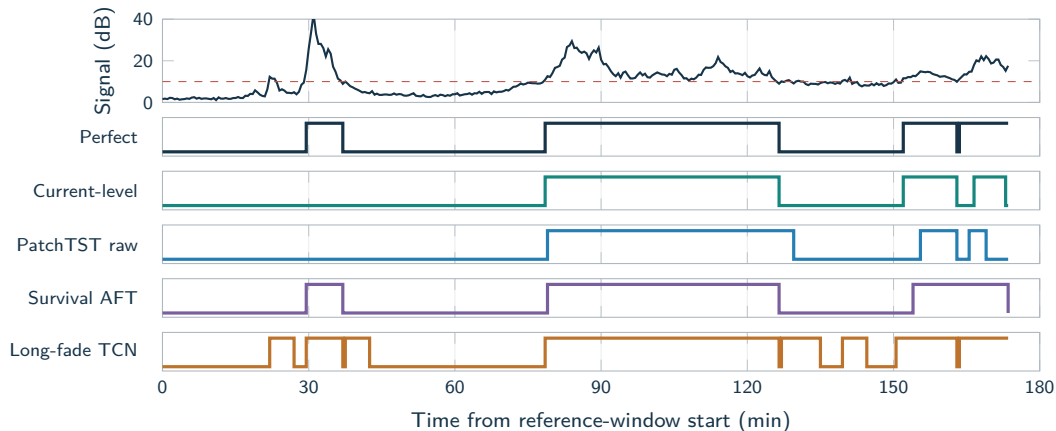
Native autoregressive grid:

Perfect: **18 intervals**

PatchTST: **20 intervals**

An interval is a contiguous run of switch 1, not necessarily a distinct physical fade. Native counts are not cross-task counts.

An Event-Level Comparison



Event dataset_007_event_00007. Final decisions: low = 0, high = 1. Missing native decisions map to 0; shared Perfect reference.

Limitations and Next Steps

Experimental limitation

Next step

One external test source

More terminals and independent acquisition periods

Exploratory survival threshold

Validation-only, cost-aware probability calibration

Offline evaluation and partial coverage

Streaming prototype with an explicit fallback policy

A deployed controller also needs a data-availability audit, persistent switch state, latency measurements and comparison against the engineering baseline.

These are offline experimental results, not evidence of deployment readiness.

Conclusions

Autoregressive	Effective trajectory-based decisions; lower forecast error does not guarantee better switches.
Current-level	Conservative XGBoost switches despite large duration errors; level recovery is not fade recovery.
Long-fade	Strong event classification, but total event duration is poorly aligned with residual switch timing.
Survival	Directly targets residual fade persistence; probability and decision-threshold calibration remain necessary.

One external holdout and an exploratory survival threshold: no universal winner established.

Evaluate target alignment, prediction quality and the final switch policy together.

Thank you

for your attention



Questions & discussion

Machine Learning for Satellite Link Fade Nowcasting

► Backup material

B1 | Exact Switch-Conversion Rules

All rules below produce the **raw binary decision** at time t .

Formulation	Activate when ...
Autoregressive	$\hat{S}_{t+k} \geq 10$ for every $k = 1, \dots, 10$.
Current-level	$S_t \geq 10$ and $\hat{R}_t^{\text{level}} \geq 300 \text{ s}$.
Long-fade	$P(D_{\text{event}} \geq 300 \text{ s} \mid X_t) \geq 0.5$.
Survival	$S_t \geq 10$ and $\hat{\mathcal{S}}(300 \text{ s} \mid X_t) \geq 0.65$.

S_t : current signal in dB. $\mathcal{S}$: survival probability. The survival threshold 0.65 is exploratory/test-selected, not validation-calibrated.

Then apply the same short-island extension and above-threshold holding rule.

B2 | Perfect Switch and Post-Processing

An offline reference

At 30-second sampling, find **11 consecutive observations** with $S \geq 10$ dB. They span ten intervals, or 300 seconds.

Mark each qualifying window. Overlapping windows recover the qualifying threshold run.

The reference sees the true future. It is an **oracle benchmark**, not a deployable predictor.

$$b_t = \begin{cases} 1, & b_{t-1} = 1 \text{ and } S_t \geq 10, \\ e_t, & \text{otherwise,} \end{cases} \quad e_t = \text{extended decision.}$$

Evaluate the held switch, not the legacy adjusted diagnostic. Ten active samples contribute 300 s to active duration.

Shared binary post-processing

- 1 Extend each short positive island to ten samples, bounded by its processing group.
- 2 Hold an active decision while the current signal remains above threshold.

This may connect nearby islands; it does **not** merge every gap or remove short islands.

B3 | Models and Input Representations

Task	Models	Input variants
Autoregressive	GRU S2V; GRU encoder–decoder; PatchTST; XGBoost	Raw and context-standardized; $L = 30$, $h = 10$.
	Chronos zero-shot	Raw; $L = 30$ and $L = 120$; $h = 10$.
Current-level	Shapelet MLP, convolution and Transformer; XGBoost	Relative-to-current context plus scalar features.
Long-fade	XGBoost; TCN; shapelet convolution	Relative-to-threshold context plus scalar features.
Survival	XGBoost-AFT; discrete-time TCN	Selected AFT: raw lags. Selected TCN: relative-to-current plus scalars.

Unless noted otherwise, context length is 30 observations. Every input feature comes from Signal history; no meteorological covariates are used.

Different outputs require different labels, but all decisions are evaluated against Perfect Switch.

B4 | Search Scope and Model Selection

Search	Trials	Validation criterion
Autoregressive GRU (two architectures, two variants)	576	RMSE ↓
Autoregressive PatchTST (two variants)	1,728	RMSE ↓
Autoregressive XGBoost (two variants)	2,304	RMSE ↓
Current-level shapelets (three heads)	648	MAE in seconds ↓
Current-level XGBoost	1,152	MAE in seconds ↓
Long-fade (three model families)	8,928	AUPRC ↑
Survival XGBoost-AFT	1,920	AFT negative log-likelihood ↓
Survival TCN	1,620	Integrated Brier score ↓

Archived grid counts across the included variants. Chronos is zero-shot. The survival TCN grid follows controlled validation experiments on representation, binning and weighting.

Hyperparameters are selected on validation; refit or checkpoint promotion is recorded in run metadata.

B5 | Normalization and Data Availability

Context standardization

$$\mu_t = \frac{1}{L} \sum_{j=0}^{L-1} s_{t-j}, \quad \tilde{X}_t = \frac{X_t - \mu_t}{s_t}$$

s_t is the context standard deviation, with a numerical safeguard. Targets use the **same context statistics**.

$$\hat{Y}_t^{\text{raw}} = s_t \hat{Y}_t^{\text{standard}} + \mu_t$$

No future-window statistics enter this transformation.

Auxiliary scalar features

Past/current signal level, window summaries and slopes; train-fitted feature scaling.

Small-gap reconstruction

Offline linear interpolation: at most two inserted samples in a gap of at most 90 s. Longer gaps separate acquisition segments.

Interpolation can require the right endpoint: an online implementation must account for its availability and latency.

An external test source is held out; development splits are chronological and event-aware.

B6 | Event Definitions and Censoring

Current-level	Search the full continuous signal for the first value below $S_t - 0.5$. A local event-window boundary does not terminate the target.
Long-fade	Group threshold crossings within three hours of the first crossing. Temporary exits between grouped crossings remain inside the event.
Survival	A state machine confirms recovery instead of splitting at every temporary threshold exit. Estimate time to the end of this event.

Observed event end

Residual duration R_t is known.
AFT bounds: $[R_t, R_t]$.

Acquisition ends first

Only follow-up C_t is known: $R_t > C_t$.
AFT bounds: $[C_t, +\infty)$.

Survival keeps right-censored samples. Ordinary current-level duration regression excludes them. A later sample across a long gap does not prove continuous persistence.

B7 | Large Duration Error, Correct Switch?

Assume the current signal is already ≥ 10 dB. Compare the duration gate at 300 s:

Illustrative example	True duration	Prediction	Absolute error
Both above 300 s	3,600 s	600 s	3,000 s
Opposite sides of 300 s	310 s	290 s	20 s

First example

Large numerical error, but the same threshold decision.

Second example

Small numerical error, but the wrong threshold decision.

Illustrations, not measured samples. Even a correct current-level duration gate is only a proxy for Perfect Switch: current-level recovery is not fade recovery.

Duration MAE, threshold accuracy and final switch quality answer different questions.

B8 | Native Grid, Reference Grid, Intersection

Evaluation domain	What it measures	Main limitation
Native task grid	Decision quality where the task produces predictions.	Different timestamp sets across tasks.
Common reference grid	One reference timeline; unavailable decisions map to 0.	Quality and native coverage are combined.
Strict intersection	Only timestamps with native decisions from all compared tasks.	Smaller, selected domain: 771 timestamps.

Strict-intersection diagnostic: switch F1

Current-level XGBoost: **0.900** Survival AFT: **0.872** Autoregressive XGBoost raw: **0.866**

These are not the reference-grid scores on the main slides. Mapping missing decisions to 0 is an evaluation convention, not a validated deployment fallback.

State the evaluation grid before interpreting a cross-task ranking.

B9 | How the Survival Models Produce Probabilities

XGBoost-AFT

$$\log R = f(X) + \tau Z$$

Trees predict a log-time location. A noise distribution and scale τ define a duration distribution:

$$\mathcal{S}(u | X) = 1 - F_Z\left(\frac{\log u - f(X)}{\tau}\right).$$

Observed durations contribute density terms; censored follow-up contributes survival terms.

Selected model: logistic noise, scale 1, raw lags, uniform sample weights.

Discrete-time causal TCN

$$q_k = P(R \text{ ends in bin } k \mid R \text{ reached bin } k, X)$$

A causal sequence encoder produces conditional event probabilities (hazards):

$$\mathcal{S}(b_k | X) = \prod_{j=1}^k (1 - q_j).$$

Training accounts for observed events and available censoring follow-up.

Selected setup: relative-to-current plus scalar inputs, hybrid bins, uniform sample weights.

Both learn from the same continuous survival labels; neither treats censored time as an observed end.

B10 | Survival Results and Statistical Uncertainty

Native external test	C-index $\uparrow$	IBS $\downarrow$	Brier@300 s $\downarrow$	Switch F1 $\uparrow$
XGBoost-AFT	0.672	0.117	0.218	0.863
Discrete-time TCN	0.658	0.103	0.214	0.784

Is the C-index difference convincing?

Paired event bootstrap: 55 external-holdout events, 2,000 resamples.

AFT minus TCN: $\Delta = 0.014$, with 95% percentile interval $[-0.021, 0.049]$.

The interval includes zero: no statistically significant ranking advantage is established. Lower TCN Brier scores do not imply better switches at one operating threshold.

Switch F1 uses the exploratory threshold 0.65 and shared post-processing. The bootstrap statement concerns the C-index difference only.

Discrimination, probabilistic accuracy and operational decisions are distinct evaluation levels.